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HISTORY · The Gupta Empire: administration, economy, art, literature

UNDERSTAND IT FIRST

What is it? After the Mauryas and a gap of about 500 years, the **Guptas** (c. 320–550 CE) again united north India from **Magadha (Pataliputra)**. This age is often called the **'Golden Age'** because of great progress in literature, science and art, but remember that the pillars of that glory (land grants, rigid castes, falling trade) also began the **feudal** pattern of the next centuries. Learn it again as **Sources → Rulers → Politics → Society → Economy → Religion → Art, literature, science → Decline**.

1. Sources

Source	What it tells
Allahabad Pillar inscription (Prayaga Prashasti)	In Sanskrit, composed by **Harishena**, court poet of **Samudragupta**. Lists his conquests. The same Ashokan pillar also carries Ashoka's edict.
Fa-Hien (Faxian)	Chinese Buddhist monk who visited India (c. 399–414 CE) in the reign of **Chandragupta II**; describes mild laws, no passports, the **chandalas** living outside towns, and prosperity.
Other inscriptions	**Mehrauli iron pillar** (Chandra = Chandragupta II); **Eran** (first sati, 510 CE); **Junagadh** (Skandagupta repaired the **Sudarshana lake**); **Bhitari** (Skandagupta and Huns); Mathura.
Coins and texts	Gold coins called **dinars** (also suvarna), silver **rupaka**, copper coins. Types: **lyrist** (Samudragupta plays the veena), **archer**, **Ashvamedha**, **tiger-slayer**. Texts: Kalidasa's works, **Devichandraguptam** and **Mudrarakshasa** (Vishakhadatta), Puranas, Narada and Brihaspati smritis.

2. The rulers, in order

Ruler	Years (CE)	Key facts
Sri Gupta, Ghatotkacha	c. 240–319	Founders of the dynasty (small rulers, titled **Maharaja**).
Chandragupta I	c. 319/320–335	First important king, title **Maharajadhiraja**. The **Gupta era** starts from his accession (319–320 CE). Married **Kumaradevi**, a **Lichchhavi** princess, which brought political strength (shown on the **Chandragupta–Kumaradevi** coins).
Samudragupta	c. 335–375	Called the **'Napoleon of India'** (V.A. Smith). The Allahabad Prashasti: in **Aryavarta** he uprooted 9 kings; in **Dakshinapatha** he captured and then released 12 kings (a policy of 'grahana-moksha-anugraha'); forest tribes and frontier states paid tribute. Performed the **Ashvamedha**; a musician (lyrist

		coins); a poet, titled **Kaviraja**. Sri Lankan king **Meghavarna** got his permission to build a monastery at **Bodh Gaya**.
Chandragupta II (Vikramaditya)	c. 375–415	Defeated the **Shakas** of Western India (Rudrasimha III), making **Ujjain** a second capital. Gave his daughter **Prabhavati Gupta** to the **Vakataka** king Rudrasena II. **Navaratnas** (nine gems) at his court, including **Kalidasa**. **Fa-Hien** visited. Title **Vikramaditya**.
Kumaragupta I	c. 415–455	Founded the **Nalanda Mahavihara** (later a great university). Performed the Ashvamedha. First Huna pressure.
Skandagupta	c. 455–467	Repelled the **Huns** (Hephthalites); repaired the **Sudarshana lake** (Junagadh inscription). After him the empire weakened: Budhagupta, later kings; the **Huns (Toramana, Mihirakula)** took north-west India. The line ended c. **550 CE**.

EXAMPLE

Andhra link (all regions equal): the Allahabad Prashasti names Samudragupta's southern opponents, identified with the Andhra–Kalinga coast and Krishna–Godavari region: **Hastivarman of Vengi** (Salankayana, near Eluru), **Ugrasena of Palakka** (Nellore area), **Kubera of Devarashtra**, **Damana of Erandapalla**, **Nilaraja of Avamukta**, **Mantaraja of Kurala**, **Vishnugopa of Kanchi** (Pallava). Some identifications are debated, but the names Vengi and Palakka are a favourite exam point.

3. Politics · Society · Economy · Religion of the Guptas

Area	Gupta period
Politics	King's titles: **Maharajadhiraja, Parama Bhattaraka**. Council of ministers; **Sandhivigrahika** (war and peace), **Mahabaladhikrita** (army), **Kumaramatya** (high officers), **Mahadandanayaka**. **Bhukti** (province, ruled by **Uparika**) → **Vishaya** (district, **Vishayapati**) → **Vithi** → **Grama** (village, **Gramika**). A **less centralised** state than the Mauryan: feudatories (**Samantas**) and land grants with administrative rights; punishments were milder; separate civil and criminal law (Narada and Brihaspati smritis). Taxes: **bhaga, bhoga, kara, udranga**, forced labour **vishti**.
Society	**Caste became more rigid**: many sub-castes; **chandalas** (untouchables) lived outside towns and beat clappers (Fa-Hien); the **Kayastha** (scribes) rose as a new group. **Status of women fell**: child marriage, the first evidence of **sati** in the **Eran** inscription (510 CE, Goparaja's wife). Shudras could listen to epics and Puranas; slavery continued.
Economy	**Largest number of gold coins** (dinars) in ancient India, but gold purity fell later. **Land grants** to Brahmins and temples (**agrahara, brahmadeya, devadana**) with rights to revenue: a start of **feudalism**. **Roman trade declined**, towns shrank, though guilds (**shrenis**), silk and ivory continued; ports **Tamralipti**, Broach. Agriculture and irrigation (Sudarshana lake).
Religion	**Brahmanism revived**: Vedic rites (Ashvamedha), **Vaishnavism** favoured (kings called **Parama Bhagavata**, emblem **Garuda**, Vishnu's **Dashavatara**), Shaivism and Shaktism grew. **Structural temples** began. **Buddhism** (Mahayana, Nalanda) and **Jainism** continued (Valabhi council c. 5th–6th century).

4. Art, architecture, literature and science

Field	Details
Temples	First **structural temples** (flat roof, then early shikhara): **Dashavatara temple, Deogarh** (stone, with Shesha-shayi Vishnu panel); **Bhitargaon** (brick, Kanpur); **Tigawa**; **Udayagiri caves** (Vidisha: Varaha panel); **Sanchi** temple 17.
Sculpture, painting	**Sarnath Buddha** (calm, plain robe, halo) and **Mathura** school; **Ajanta** paintings (caves 16, 17, 1: Jataka stories, Padmapani); **Bagh caves**. Metal work: **Mehrauli iron pillar** and the bronze Buddha of Sultanganj.
Literature	**Kalidasa**: *Abhijnana Shakuntalam*, *Meghaduta*, *Raghuvamsha*, *Kumarasambhava*, *Ritusamhara*, *Malavikagnimitra*, *Vikramorvasiya*. **Shudraka**: *Mrichchhakatika*. **Vishakhadatta**: *Mudrarakshasa*. **Amarasimha**: *Amarakosha*. **Vishnu Sharma**: *Panchatantra*. **Vatsyayana**: *Kamasutra*. Languages: Sanskrit as court language, Prakrit for common speech in plays.
Science	**Aryabhata**: *Aryabhatiya* (499 CE), the place-value system and decimal, value of pi about 3.1416, earth rotates on its axis, causes of eclipses. **Varahamihira**: *Brihat Samhita*, *Panchasiddhantika*. **Vagbhata**: *Ashtanga Hridaya* (medicine). **Charaka** and **Sushruta** samhitas were compiled earlier. **Nalanda** was a centre of learning.

5. Decline

Causes: (1) Hun invasions (Toramana, Mihirakula); (2) weak later rulers and quarrels; (3) growth of powerful feudatories (Samantas) and land grants that drained revenue; (4) decline of trade and towns, less money; (5) rise of regional powers: Vakatakas, Maukharis, Later Guptas, Yashodharman of Malwa, Pushyabhutis.

EXAM TRAP

Allahabad Pillar: Harishena (Samudragupta); Mehrauli iron pillar: **Chandra**; **Fa-Hien** under **Chandragupta II**; **Nalanda: Kumaragupta I**; **Huns repelled: Skandagupta**; sati at **Eran (510 CE)**; **Aryabhatiya (499 CE)**; Brihat Samhita: **Varahamihira** (not Aryabhata); Mudrarakshasa: **Vishakhadatta**. Gupta rule was **less centralised** than Mauryan. Gupta gold coins are called **dinars**; Mauryan silver coins were **punch-marked panas**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the Gupta period:

1. The Allahabad Pillar inscription, composed by Harishena, describes the conquests of Samudragupta.

2. The Chinese monk Fa-Hien visited India in the reign of Chandragupta II.

3. Chandragupta I married Kumaradevi, a Lichchhavi princess.

4. Skandagupta founded the Nalanda Mahavihara.

Which of the above are correct?

A. 1, 2 and 3 only

B. 1, 3 and 4 only

C. 2 and 4 only

D. 1, 2, 3 and 4

Answer: A. Nalanda is associated with Kumaragupta I, so 4 is wrong.

PRACTICE

Match List-I (ruler) with List-II (feature):

List-I (ruler)	List-II (feature)
A. Samudragupta	1. Defeated the Shakas; Ujjain a second capital
B. Chandragupta II	2. Repelled the Huns; Sudarshana lake repaired
C. Kumaragupta I	3. Allahabad Prashasti; lyrist coins
D. Skandagupta	4. Founded Nalanda

(1) A → 3 B → 1 C → 4 D → 2

(2) A → 1 B → 3 C → 4 D → 2

(3) A → 3 B → 1 C → 2 D → 4

(4) A → 3 B → 4 C → 1 D → 2

Answer: (1) Samudragupta: Prashasti and lyrist coins; Chandragupta II: Shakas; Kumaragupta I: Nalanda; Skandagupta: Huns and Sudarshana lake.

PRACTICE

Which one of the following statements about the Gupta period is NOT correct?

A. The Guptas issued the largest number of gold coins in ancient India.

B. The first epigraphic evidence of sati comes from the Eran inscription of 510 CE.

C. Land grants to Brahmins with revenue rights became common.

D. Aryabhata wrote the Brihat Samhita.

Answer: D. Brihat Samhita was written by Varahamihira; Aryabhata wrote the Aryabhatiya.

PRACTICE

Assertion (A): Gupta administration was less centralised than the Mauryan administration.
Reason (R): The Guptas made land grants with administrative rights and depended on feudatories (Samantas).
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Land grants and feudatories spread power away from the centre.

PRACTICE

A Chinese Buddhist monk who came to collect Buddhist texts wrote that people travelled without passports, punishments were mild and untouchables struck wooden clappers while entering towns. In whose reign did he visit?

A. Chandragupta II (Vikramaditya)

B. Ashoka Maurya

C. Harshavardhana

D. Kanishka

Answer: A. This is Fa-Hien's account, c. 399–414 CE, in Chandragupta II's reign. (Harsha's visitor was Xuanzang.)

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the Gupta Politics–Society–Economy–Religion table from memory. 2. Write the 6 rulers with one key fact each. 3. List 5 literature and 4 science names with their works.

POLITY · Fundamental Rights II: Articles 20–22, 23–24, 25–30, 32 and writs

UNDERSTAND IT FIRST

What is it? Day 4 covered equality and the six freedoms (Arts 14–19). Today the rest of Part III: **protection in criminal trials and life and liberty (Arts 20–22)**, **no exploitation (23–24)**, **freedom of religion (25–28)**, **minorities (29–30)** and the **remedy: the right to go to court (Art 32)**. Think of Art 32 as the 'lock' that protects all the other rights.

1. Right to Freedom (continued): Articles 20, 21, 21A, 22

Article	What it says	Key points and cases
Art 20	Protection in respect of conviction for offences: (1) **no ex-post-facto law** (no punishment for an act that was not an offence when done, and no higher penalty than the law then in force); (2) **no double jeopardy** (not prosecuted and punished twice for the same offence); (3) **no self-incrimination** (nobody forced to be a witness against himself).	Available to **all persons** (citizens and foreigners), only in **criminal** proceedings. **Cannot be suspended** even during an emergency (Art 359, after the 44th Amendment).
Art 21	**Protection of life and personal liberty**: no person shall be deprived of his life or personal liberty except according to **procedure established by law**.	**A.K. Gopalan (1950):** narrow reading. **Maneka Gandhi (1978):** the procedure must be **just, fair and reasonable**; Arts 14, 19 and 21 are connected, so Art 21 became a source of many rights: **privacy** (**K.S. Puttaswamy, 2017**), **livelihood** (**Olga Tellis, 1985**), **speedy trial** (**Hussainara Khatoon, 1979**), clean environment, dignity, shelter, health, free legal aid, right to die with dignity (passive euthanasia: Common Cause 2018). Applies to **all persons**; cannot be suspended in emergency.
Art 21A	**Right to education**: free and compulsory education for children aged **6–14**.	Added by the **86th Amendment (2002)**; the **RTE Act 2009** came into force on **1 April 2010**.
Art 22	**Protection against arrest and detention. Ordinary arrest:** must be told the grounds, right to consult and be defended by a lawyer, and must be produced before a **magistrate within 24 hours** (excluding travel time).	These protections **do not apply** to **enemy aliens** and to **preventive detention** (detaining to prevent a future crime). Preventive detention up to **3 months** without the opinion of an **Advisory Board** (the 44th Amendment's change to 2 months is not yet in force). Laws: PD Act 1950, MISA 1971, COFEPOSA 1974, **NSA 1980**, UAPA. **D.K. Basu (1997)**: arrest guidelines.

2. Right against Exploitation: Articles 23 and 24

Article	Meaning	Notes
Art 23	Prohibits **traffic in human beings, begar (unpaid**	Available even against **private persons**. Exception: the State may impose **compulsory service for public purposes** (e.g., military or national

	forced labour) and other similar forced labour.	service). Laws: **Immoral Traffic (Prevention) Act 1956**, **Bonded Labour System (Abolition) Act 1976**. **PUDR v. Union of India (1982, Asiad workers):** paying less than minimum wages is forced labour.
Art 24	No child **below 14** to work in any **factory, mine or other hazardous employment**.	**Child Labour (Prohibition and Regulation) Act 1986**, amended in **2016** (ban on all work below 14, with exceptions for family work; adolescents 14–18 barred from hazardous work). **M.C. Mehta v. State of Tamil Nadu (1996):** Sivakasi match factories.

3. Right to Freedom of Religion: Articles 25–28

Article	Meaning	Notes
Art 25	**Freedom of conscience** and the right to **profess, practise and propagate** religion, to **all persons**.	Subject to **public order, morality, health**; the State can regulate secular activity linked to religion and provide social reform (e.g., temple entry for all Hindu classes). Wearing and carrying the **kirpan** is part of the Sikh religion. **Shirur Mutt (1954):** 'essential religious practices' test.
Art 26	Every **religious denomination** may establish and maintain institutions, **manage its own affairs in religion**, own and administer property.	Right of a **group** (denomination), not an individual.
Art 27	No person can be **forced to pay taxes** whose proceeds go to promote or maintain any **particular religion**.	Fees are allowed; taxes are not.
Art 28	**No religious instruction** in institutions **wholly maintained by the State**; in aided or recognised institutions it is voluntary.	**Bijoe Emmanuel (1986):** students may refuse to sing the national anthem on religious grounds, but must not disrespect it. **S.R. Bommai (1994):** secularism is part of the basic structure.

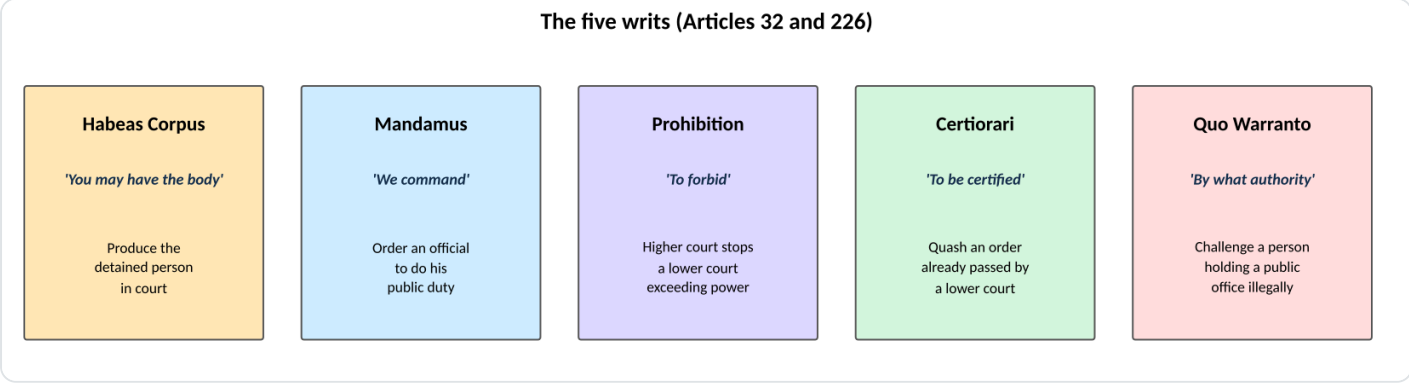
4. Cultural and Educational Rights: Articles 29–30

Article	Meaning	Notes
Art 29	Any section of **citizens** with a **distinct language, script or culture** has the right to **conserve** it. **No citizen can be denied admission** to a State-run or State-aided institution **only on grounds of religion, race, caste or language**.	Protects both **minorities and majority** sections (only for citizens).
Art 30	All **minorities, based on religion or language**, have the right to **establish and administer educational institutions** of their choice.	State cannot discriminate in giving aid on this ground. If the State acquires such property, it must pay **compensation (44th Amendment)**.

		T.M.A. Pai Foundation (2002): minority status is decided **State-wise**.

5. Right to Constitutional Remedies: Article 32 and the writs

Art 32 gives every person the right to move the **Supreme Court** directly to enforce Fundamental Rights. Dr. **Ambedkar** called it the **'heart and soul'** of the Constitution. The Court can issue **directions, orders or writs**. Art 32 itself is a Fundamental Right; it can be suspended only during an emergency (Art 359). **High Courts** have the same power under **Art 226**, which is **wider**: they can issue writs for FRs **and for any other legal right**. Writs are an idea taken from **English law**.



The five writs (Articles 32 and 226)

Writ	Meaning	When used
Habeas Corpus	'You may have the body'	Court orders the person who has detained someone to **produce him in court** so that the legality of detention can be tested. Can be used against **public authorities and private persons**. Protects personal liberty.
Mandamus	'We command'	Orders a **public official or body** to do a **public legal duty** they have refused to do. **Not** issued against private individuals (generally), the **President or Governors**, or for a duty that is only discretionary.
Prohibition	'To forbid'	A **higher court** forbids a **lower court or tribunal** from going **beyond its jurisdiction** while a case is still pending. Used against judicial and quasi-judicial bodies only.
Certiorari	'To be certified'	A higher court **quashes an order already passed** by a lower court or tribunal for want of jurisdiction or an error of law. (Prohibition is **before** the decision; certiorari is **after**.)
Quo Warranto	'By what authority'	Court asks a person **by what authority he holds a public office** and removes him if he holds it illegally. Only for a **substantive public office created by law** (not private or ministerial posts).

Other articles of Part III: Art 33 lets Parliament restrict FRs of the **armed forces and police**; **Art 34** limits FRs while **martial law** is in force; **Art 35** gives Parliament (not States) the power to make laws for certain FRs. **ADM Jabalpur (1976)** held that habeas corpus could not be used in the Emergency, but it was overruled in **Puttaswamy (2017)**.

EXAMPLE

Example: Ravi is arrested and kept in a police station for 3 days without being shown to a magistrate. This breaks **Art 22(2)** (24 hours). His family can ask the High Court or the Supreme Court for a **habeas corpus** (Art 226 or 32) to get him produced. If a man holds the post of a public official illegally, a **quo warranto** petition challenges his authority; if an order of a tribunal exceeded its powers, **certiorari** can cancel it.

EXAM TRAP

Arts 20 and 21 cannot be suspended in an emergency. **Art 20** is for **criminal** matters only. **Art 22** protections **do not apply to preventive detention**. **Art 21A**: 86th Amendment, ages **6–14**. **Habeas corpus**: against private persons too; **mandamus**: not against private persons, President or Governors. **Prohibition = before, certiorari = after. Quo warranto**: public office only. **Art 32** = Supreme Court; **Art 226** = High Court (wider). **Art 29** can be used by any section of citizens; **Art 30** by minorities.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about Article 20 of the Constitution:
1. It protects against ex-post-facto criminal laws.
2. It protects against double jeopardy.
3. It cannot be suspended during a national emergency.
4. It is available only to citizens.
Which of the above are correct?

A. 1, 2 and 3 only

B. 1 and 4 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. Art 20 is available to all persons, citizens and foreigners, so 4 is wrong.

PRACTICE

Match List-I (writ) with List-II (meaning/purpose):

List-I (writ)	List-II (meaning/purpose)
A. Habeas Corpus	1. 'By what authority'
B. Mandamus	2. 'We command'
C. Certiorari	3. Quash an order already passed by a lower court

D. Quo Warranto	**4.** 'To have the body'

(1) A → 4 B → 2 C → 3 D → 1

(2) A → 2 B → 4 C → 3 D → 1

(3) A → 4 B → 2 C → 1 D → 3

(4) A → 3 B → 2 C → 4 D → 1

Answer: (1) Habeas corpus: have the body; mandamus: we command; certiorari: quash a passed order; quo warranto: by what authority.

PRACTICE

Which one of the following statements about Article 21 is NOT correct?

A. In Maneka Gandhi v. Union of India (1978) the Court held that the procedure must be just, fair and reasonable.

B. The right to privacy was held to be part of Article 21 in K.S. Puttaswamy (2017).

C. Article 21A guarantees free and compulsory education to children aged 6–14.

D. Article 21 is available only to citizens.

Answer: D. Article 21 protects all persons, including foreigners.

PRACTICE

Assertion (A): A writ of habeas corpus can be issued even against a private person.
Reason (R): The aim of the writ is to free a person who is illegally detained, whoever has detained him.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Habeas corpus protects personal liberty against illegal detention by anyone, including private persons.

PRACTICE

A person is arrested and kept in police custody for three days without being produced before a magistrate. Which provision is violated and which writ is most suitable?

A. Article 22(2); habeas corpus

B. Article 20(3); mandamus

C. Article 19(1)(d); quo warranto

D. Article 23; certiorari

Answer: A. Art 22(2) requires production before a magistrate within 24 hours; habeas corpus tests the legality of detention.

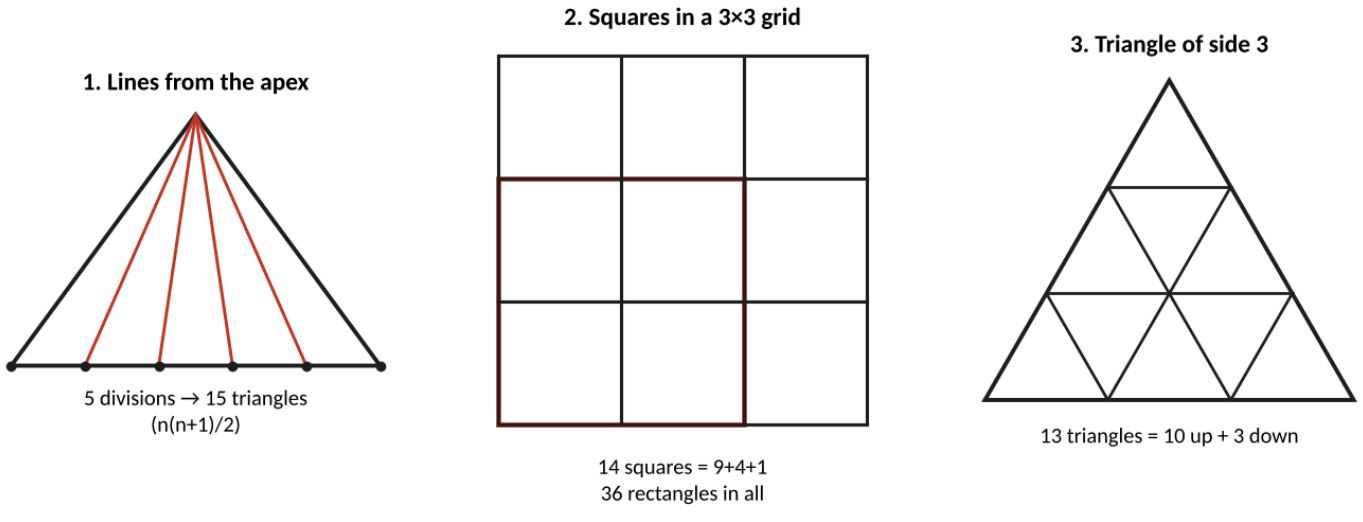
RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write Arts 20–22 with the three protections of Art 20, the Art 21 cases and the Art 22 rules. 2. Write Arts 23–30 in one line each. 3. Draw the five writs with meaning and when used; write the difference between Art 32 and Art 226.

REASONING · Shapes and their sub-sections (figures and counting)

UNDERSTAND IT FIRST

What is it? A figure (a triangle with many lines, a grid of squares, a polygon) is given, and you must **count how many triangles, squares or rectangles it contains, of every size**. **The secret:** never count by eye. Count **systematically**: name the points with letters, count the **smallest pieces first, then pieces made of 2, 3 … small pieces**, and use a formula to check.



Three standard counting figures

Case	Formula	Example
Triangles with lines from the apex to n divisions of the base	n(n+1)/2	n = 5: 5×6/2 = **15**
Line segments on a line with n points	n(n−1)/2	6 points: 6×5/2 = **15**
Diagonals of an n-sided polygon	n(n−3)/2	Octagon: 8×5/2 = **20**; hexagon: **9**
Rectangles (including squares) in an m×n grid	m(m+1)×n(n+1)/4	3×3: 3×4×3×4/4 = **36**; 3×4: **60**
Squares in an n×n grid	$1^2 + 2^2 + \ldots + n^2 = n(n+1)(2n+1)/6$	3×3: 14; 4×4: 30; chessboard 8×8: **204**
Triangle of side n cut by lines parallel to its sides (n = 3)	Up-pointing + down-pointing	6 + 3 + 1 = 10 up; 3 down; total **13** (n = 4: **27**)

EXAMPLE

Steps for any figure: (1) letter the corners; (2) count **single** small figures; (3) count figures made of **2** parts, then **3** parts and so on; (4) add; (5) check with a formula if one fits. In a triangle with lines from the apex, every triangle is fixed by choosing **two points on the base**: with 6 points that is 6×5/2 = 15 pairs.

EXAM TRAP

Do not forget **bigger figures made of several small pieces** and, in squares, the **tilted** squares if the figure shows diagonals. In a grid, rectangles **include squares**; if the question says 'rectangles that are not squares', subtract the squares (3×3 grid: 36 − 14 = 22).

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

The base of a triangle is divided into 5 equal parts, and lines are drawn from the apex to each division point. How many triangles are there in the figure?

A. 10

B. 15

C. 20

D. 21

Answer: B. n(n+1)/2 = 5×6/2 = 15 (choose 2 of 6 base points).

PRACTICE

How many squares of all sizes are there in a 4×4 grid of small squares?

A. 16

B. 20

C. 30

D. 36

Answer: C. 16 + 9 + 4 + 1 = 30.

PRACTICE

How many rectangles (including squares) can be counted in a grid made of 3 rows and 4 columns of small squares?

A. 40

B. 50

C. 60

D. 72

Answer: C. (3×4/2) × (4×5/2) = 6 × 10 = 60.

PRACTICE

An equilateral triangle of side 3 units is divided by lines parallel to its sides into 9 small equilateral triangles of side 1 unit. How many triangles of all sizes are there in the figure?

A. 9

B. 11

C. 13

D. 15

Answer: C. Up-pointing: 6 (size 1) + 3 (size 2) + 1 (size 3) = 10; down-pointing: 3 (size 1). Total 13.

PRACTICE

How many diagonals does a regular octagon have?

A. 16

B. 18

C. 20

D. 28

Answer: C. n(n−3)/2 = 8×5/2 = 20.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the six formulas and one example for each. 2. Draw a 3×3 grid and count squares by size (1×1, 2×2, 3×3). 3. Draw a triangle with 4 lines from the apex and count triangles by hand and by formula.

NOTEBOOK PAGE · Day 5: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

HISTORY: Guptas, Politics · Society · Economy · Religion

Area	Write
Sources	Allahabad Pillar (Harishena), Mehrauli iron pillar, Fa-Hien, Eran (sati 510 CE), Junagadh (Skandagupta), coins (dinars)
Rulers	Chandragupta I (320, Kumaradevi); Samudragupta (Allahabad, Dakshinapatha, lyrist, Ashvamedha); Chandragupta II (Shakas, Ujjain, Fa-Hien, Navaratnas, Prabhavati); Kumaragupta I (Nalanda); Skandagupta (Huns, Sudarshana); last c. 550
Politics	Bhukti → Vishaya → Vithi → Grama; Uparika, Vishayapati, Kumaramatya, Sandhivigrahika, Mahabaladhikrita; Samantas and land grants = early feudalism; mild punishments
Society	Rigid castes, chandalas outside; Kayasthas; women's status lower, sati (Eran), child marriage; Fa-Hien: vegetarian, mild
Economy	Gold dinars (many), silver, copper; land grants (agrahara); decline of Roman trade, towns decline; guilds; agriculture
Religion	Brahmanism and Vaishnavism revived (Garuda, Dashavatara); structural temples (Deogarh, Bhitargaon); Buddhism (Nalanda), Jainism continued
Science, literature, art	Kalidasa, Shudraka, Vishakhadatta, Amarasimha, Vishnu Sharma, Vatsyayana; Aryabhata, Varahamihira, Vagbhata; Ajanta, Sarnath Buddha, iron pillar

POLITY: Fundamental Rights II

- Art 20 (3 protections; not suspended), Art 21 (procedure established by law; Maneka Gandhi; Puttaswamy; 21A), Art 22 (24 hours; preventive detention 3 months).
- Art 23 (traffic in humans, begar), Art 24 (below 14, hazardous); Arts 25–28 (religion), Arts 29–30 (minorities); Art 32 and the 5 writs with meanings; Art 226.

REASONING: Figures and counting

Write the formulas: triangles from apex n(n+1)/2; line segments n(n−1)/2; diagonals n(n−3)/2; rectangles in m×n grid = m(m+1)n(n+1)/4; squares = Σ(m−k+1)(n−k+1). Draw the 3 figures and count systematically (small → big).